# Track 2 — Technical Documentation

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## Track 2 — Technical Documentation

**MVA Hackathon 2026 · Rare Disease, Real Kid · BUB1B / Mosaic Variegated Aneuploidy**

This document describes how the Track-2 deliverables in this repository are produced and verified: the evidence workflow, the data sources and APIs it draws on, the native-image verification harness, the figure set, the check harness, and the candidate-table contract. It is the engineering companion to `TRACK2_REPORT.md`; it asserts no new science.

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### 1. Purpose and scope

Track 2 of the challenge asks for a mechanism-grounded drug-reprioritization dossier for the Track-1 diagnosis (biallelic BUB1B / Mosaic Variegated Aneuploidy). The deliverables in this repository are:

| Artifact | Path | Role |
|---|---|---|
| Professional report | `TRACK2_REPORT.md`, `TRACK2_REPORT.pdf` | The submission document: mechanism characterization, therapeutic landscape, exclusion jurisprudence, six-assay validation plan |
| Lay report | `TRACK2_LAY_REPORT.md`, `.pdf` | Plain-language companion for the family/foundations audience |
| Candidate table | `candidates.tsv` | 20-row machine-readable candidate register |
| Report figures | `figures/` | F1–F5 committed renders |

| Artifact | Path | Role |
|---|---|---|
| Evidence pack | evidence/ | 19 analytical memos — 4 wave-2 + 5 wave-3 + 4 wave-4 + 3 literature scans + 3 mechanism memos, exactly as enumerated in evidence/MEMO_MANIFEST.md — plus the four adversarial review passes, adjudication memo, and structural-validation records |
| Native images | evidence/native_images/ | Third-party structural imagery with a full verification chain (§4) |
| Citation check | checks/check_citations.py | Reference-consistency gate (§6) |

**Non-goals.** Track-1 (variant calling/ranking) lives in its own repository. This document changes no scientific claim; where it quotes numbers, each is traceable to a repository artifact named next to it.

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## 2. Workflow architecture

The dossier was produced by a generation → adversarial-review → integrate loop, documented in evidence/WORKFLOW_CONSENSUS.md (the session log) and closed per evidence/SCOPE_DECISION.md:

1. **Wave-2 axis generation** — parallel research memos per mechanism axis (e.g. wave2_StabilizationAxis.md, wave2_SenolyticAxis.md, wave2_CINSyndromeAxis.md, wave2_NonsenseAxis.md), each an evidence memo.
2. **Adversarial passes** — an out-of-family reviewer attacked each integrated pack (wave2_ADVERSARIAL_REVIEW.md, wave3_ADVERSARIAL_PASS2.md, wave4_ADVERSARIAL_PASS3.md, wave5_ADVERSARIAL_PASS4.md); findings were fixed at the controller layer. Major-count decay per SCOPE_DECISION.md: **8 → 2 → 2 → 0** across the four passes.
3. **Closure** — pass 4 returned zero majors; the stop rule (two consecutive zero-major passes, second over fresh axes) was declared met: *"CONVERGENCE OUTCOME (2026-09-05): decay 8 → 2 → 2 → 0; pass 4 returned ZERO majors; workflow CLOSED-WITH-RESIDUAL-RISK."* Residuals are declared in SCOPE_DECISION.md (R1: no pass 5 exists).
4. **Contested anchors** — claims a wave could not settle received dedicated adjudication memos (e.g. adjudication_rapalog_contested_anchor.md).

The consensus matrix and candidate register are the *output* of this loop: candidates.tsv carries a consensus_status column (INCLUDE / INCLUDE-AS-METABOLIC-CONTEXT / EXCLUDE …) that the report's §4 (therapeutic landscape) mirrors; the indicators documented there are what the six-assay validation plan (report §5) operationalizes.

**Attack map.** evidence/TRACK2_ATTACK_MAP.md lists, per report section, which claims were targeted and how each attack resolved — the audit surface for a judge.

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## 3. Data sources and APIs

All external data enters through named, citable endpoints. Exact URL patterns live in evidence/native_images/fetch_native.sh.

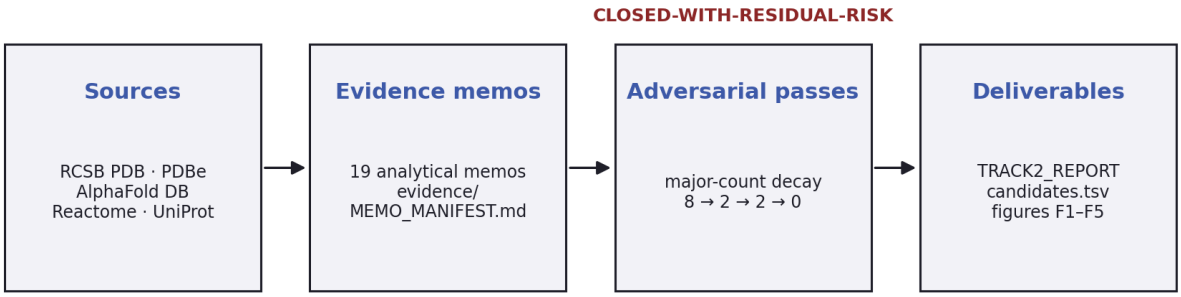



Figure 1: Workflow overview

| Source | What we use | Endpoint pattern | Docs |
|---|---|---|---|
| **Reactome** | Native pathway diagrams (R-HSA-68877, R-HSA-69620, R-HSA-5663220); flagged export with `flg=O60566`; diagram layout JSON; participants membership | `ContentService/exporter/diagram/{stId}.png`, `download/current/diagram/{stId}.json`, `ContentService/data/participants/{stId}` | reactome.org/dev/content-service |
| **RCSB PDB** | Assembly images for 6JKK, 6JKM | `https://cdn.rcsb.org/images/structures/{pdb}_assembly-1.jpeg` | rcsb.org/docs/programmatic-access |
| **PDBe** | Deposited-chain statics for 6JKK, 6JKM | `https://www.ebi.ac.uk/pdbe/static/entry/{pdb}_deposited_chain_front_image-200x200.png` | ebi.ac.uk/pdbe |
| **AlphaFold DB** | AF-O60566-F1 model + per-residue confidence JSON | `https://alphafold.ebi.ac.uk/files/AF-O60566-F1-confidence_v6.json` | alphafold.ebi.ac.uk/api-docs |
| **UniProt** | O60566 (BUB1B) feature table cited in report captions | REST API | uniprot.org/help/api |

Key facts established through these sources (all recorded in `evidence/native_images/PROVENANCE.md`):

- The Reactome exporter is **deterministic**: two unflagged fetches of the same diagram are byte-identical (negative control, 2026-09-24).
- The participants endpoint lists uniprot:**O60566** (BUB1B) as a component of **7 / 4 / 7 complexes** across R-HSA-68877 / R-HSA-5663220 / R-HSA-69620 respectively — because BUB1B is a kinetochore component in those diagrams, never a standalone node (except on R-HSA-69620, where the `BUB1B` particle, `hBUBR1:hBUB3:MAD2*:CDC20` MCC and `MCC:APC/C` are drawn explicitly).
- The core SAC sub-pathways (R-HSA-141430/141444/141405, R-HSA-69618) carry `hasDiagram=false`; R-HSA-69620 is the diagrammed parent covering the arm.

Analysis libraries: matplotlib, numpy Pillow, scipy.ndimage — no hand-rolled statistics (binomial test via `scipy.stats.binomtest`).

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## 4. Native-image verification chain

The structural imagery is not decorative: its claims are verified in three steps, each independently checkable from committed artifacts. The full text — including the negative control, the two *unexplained* R-HSA-68877 clusters, and the withdrawn earlier wording — is `evidence/native_images/PROVENANCE.md`.

1. **Fetch** (`fetch_native.sh`) — non-destructive by default (`OUT` override for refresh), records sources.
2. **Flag-vs-plain diff maps** — exported with `flg=O60566`; determinism control proves diffs are real render changes, not timestamps.
3. **Grounding** — (a) participants membership (7/4/7 complexes, peDbIds listed); (b) coincidence of diff clusters with the verified complex nodes, priced against a null baseline: 1000 random 150-unit boxes, seed 42 (`null_baseline.py`) — R-HSA-69620: **12/12** vs 9.9 % baseline, p ≈ 9e-13; R-HSA-5663220: **4/4** vs 38.8 %, p = 0.023; R-HSA-68877: **8/10** vs 58.6 %, p = 0.15 — underpowered there, so membership (3a) is the anchor on that diagram.
4. **Readable zoom set** — `make_readable.py` builds overview + zoom sheets (`readable_*_overview/zoom.png`); crops from diff clusters, node names from diagram-JSON intersection. All six committed PNGs are **byte-identical** to regeneration from the archived inputs (`python3 make_readable.py <stId> <tmpdir>`, compared with `cmp`).

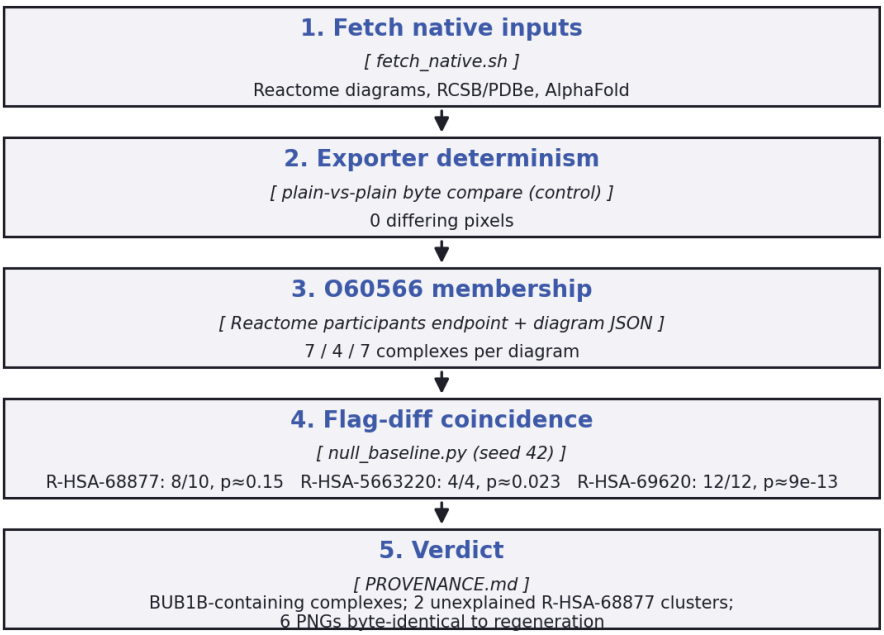



Figure 2: Verification chain

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## 5. Report figure set F1–F5

Committed renders in `figures/`; the report's captions (§2–§5) name each figure's primary sources (UniProt O60566 feature table, Ensembl MANE transcript, AlphaFold model, 6JKK crystal structure).

| Figure | Subject |
|---|---|
| F1_variant_structure.png | p.Asn1002Lys structural context (6JKK-based) |
| F2_BubR1_protein_map.png | BUBR1 functional-domain map with the two patient alleles |
| F3_BUB1B_gene_exon17.png | Exon-17 architecture for the ASO-skipping program |
| F4_decision_instrument.png | Gate instrument for therapeutic decisions |
| F5_consensus_process.png | The wave/consensus process summarized for the report |

Related structural artifacts: `evidence/6JKK.cif`, `evidence/6JKK.pdb`, `evidence/AF_BUB1B.pdb`, `evidence/plip_6jkk/` (PLiP interaction profile outputs; PLiP = Protein–Ligand Interaction Profiler).

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## 6. Citation check harness

`checks/check_citations.py` is the reference-consistency gate for the professional report:

```
python3 checks/check_citations.py $PWD    # run from anywhere; arg = repo root
```

It censuses the report's `[E#]` register (defined in report §8) against in-text citations and prints `CENSUS OK` on success. The script recomputes the numbers itself on every run; last run 2026-10-19: **77 base entries + 2 sub-rows (79 cited)**. Do not confuse this with the separate "90/90 registry-verified citations" figure in the report (32+34+24 across the three waves; per-wave literature-identifier verification, recorded in `evidence/WORKFLOW_CONSENSUS.md`) — different ledger, different count. A failing census blocks ship — run it after any edit to `TRACK2_REPORT.md`.

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## 7. Candidate-table contract

`candidates.tsv`: TSV, header + **20 data rows**.

| Column | Content |
|---|---|
| `candidate` | Therapy candidate (generic name or class) |
| `axis` | Mechanistic axis letter (proteostasis, senolysis, immune surveillance, …) |
| `MoA` | Mechanism of action, one sentence |
| `approval_status` | Regulatory positioning (e.g. "investigational; no approved indication") |
| `evidence_grade` | Letter grade (A/B/C × human/preclinical) |
| `supporting_n`, `contrasting_n` | Study counts pro/contra |
| `key_PMIDs_DOIs` | Pinned primary sources, `PMID\|DOI; …` |

| Column | Content |
|---|---|
| `consensus_status` | Wave-loop verdict (INCLUDE / INCLUDE-AS-... / EXCLUDE + reason) |

(A trailing run of empty reviewer columns `0..8` was removed on 2026-10-19: nothing consumed them and the populated content lives in `consensus_status`.)

The TSV and report §4 vote from the same consensus matrix; divergence between them is a defect — the attack map tracks any found instance.

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## 8. Reproduction

```
# native-image inputs (non-destructive; use OUT=/tmp/x to refresh)
evidence/native_images/fetch_native.sh

# verification & derived figures
python3 evidence/native_images/null_baseline.py            # prints baselines + p-values
python3 evidence/native_images/make_readable.py R-HSA-68877 /tmp/out

# doc figures (this file's two diagrams)
python3 docs/figures/build_figs.py

# report consistency gate
python3 checks/check_citations.py $PWD

# PDF of this document (pandoc, xelatex)
docs/build_pdf.sh
```

pandoc manual governs PDF options; the two `docs/figures/*.png` are regenerated by `docs/figures/build_figs.py` (matplotlib, deterministic).

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## 9. References


- Reactome pathway database — Gillespie et al., 2022, PMID 34788843.
- AlphaFold Protein Structure Database / prediction — Jumper et al., *Nature* 2021, PMID 34265844.
- PLiP (Protein–Ligand Interaction Profiler) — Salentin et al., *Nucleic Acids Res.* 2015, PMID 26089450.
- RCSB Protein Data Bank — rcsb.org/docs/programmatic-access.
- PDBe — ebi.ac.uk/pdbe.
- UniProt — uniprot.org/help/api.
- matplotlib — Hunter, *Comput. Sci. Eng.* 2007, (matplotlib.org).
- Scientific-report citation block: see `TRACK2_REPORT.md` §8 (61+2 entries, registry-verified per §3 of that report).